Charan Kumar Nara

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SUMMARY

Software Engineer with **3+** years of experience building scalable backend systems, AI/ML pipelines, and full-stack applications across fintech and health-tech domains. Proficient in Java (Spring Boot), Python (FastAPI, Flask), and JavaScript (React.js), with hands-on expertise in microservices, cloud deployments (AWS, Azure), and CI/CD automation. Experienced in integrating LLMs and vector databases into RAG systems using Bedrock, LangChain, and FAISS. Strong understanding of system design, secure APIs, data pipelines, and Agile development practices.

EDUCATION

University at Buffalo, SUNY — **Dec 2024**

Master of Science, Computer Science and Engineering — **GPA: 3.77/4.0**

- **Relevant Coursework:** Object-Oriented Languages, Data-Intensive Computing, Databases, Operating Systems, Machine Learning, Deep Learning, Computer Vision, Data Mining, Cloud Systems, Algorithms, Data Structures, Web Development, Big Data
- **Involvement:** AI & ML Club, Computer Vision Research Group, Speaker and Event Manager at CSE and Data Science Workshops

WORK EXPERIENCE:

Software Development Engineer | *Fisglobal, US* — **Aug 2024-Present**

- Engineered a low-latency fraud detection **microservice** using **Java + Spring Boot**, consuming **high-throughput Kafka streams**, invoking real-time **ML models**, and flagging anomalies – resulting in a **15% drop in fraud-related incidents**.
- Architected an enterprise **RAG system** for querying complex financial compliance documents, **integrating AWS Bedrock** and **FAISS** vector stores. Exposed the service via a secure **Spring Boot API,** cutting internal query times by over 40%.
- Built the **backend for a stakeholder analytics dashboard** using **Spring for GraphQL**, aggregating data from fraud alerts, system health, and microservices — reduced client data-fetching calls by **60%** vs. legacy REST architecture.
- Automated the **CI/CD pipeline** using **Jenkins** and **Docker** for the containerization and deployment of **Java microservices** and **AI** components, ensuring **high availability** and **consistent**, **reliable environments**.

Software Engineer Intern | *Silicon Valley AI health tech startup, US* — **Jun 2024-Oct 2024**

- Engineered real-time, scalable web application for exercise tracking using Google's **MediaPipe**, **React.js**, **Flask**, and modular architecture; improved **AI-driven pose estimation** accuracy by 20% through optimized algorithms and componentized design.
- Optimized repetition logic using **data structures** and **object-oriented design**; enhanced collaboration via **Git** version control, applied **SCRUM** best practices, and improved delivery velocity by 15% through structured task breakdown and clean code reviews.

Full Stack Software Development Engineer | *Cognizant, India* — **Oct 2021-July 2023**

- Developed a scalable **crisis communication dashboard** using **React.js** and **Next.js** accelerating communication workflows by 30% and demonstrating expertise in **large-scale system design**.
- Spearheaded fault-tolerant, low-latency microservices using **Node.js** and **AWS Lambda**, improving system response times by 20% through efficient data handling and a streamlined serverless architecture with **API Gateway**.
- Implemented a **Redis read-through caching layer** with TTL on top of PostgreSQL, cutting average read latency by 30% under peak load, tuned indexes and connection pooling to sustain high-traffic workloads.
- Orchestrated the **CI/CD** pipeline architecture with **Jenkins** and **Docker**, automating **REST API** deployments for **hardware process automation**, reducing deployment times by **40%**, and ensuring stable, consistent environments across all stages.
- Collaborated with **cross-functional teams** to manage project priorities, deadlines, and deliverables, demonstrating versatility in full-stack development while ensuring high-quality code and performance.

SKILLS

Languages	Python, C/C++, C#, Java, JavaScript/TypeScript
Frontend & UI	React.js (Redux), HTML5/CSS3, Tailwind CSS
Backend Frameworks	Spring Boot, FastAPI, Flask, Node.js, Express.js, REST APIs, GraphQL, Microservices, Django, .NET Framework (professional training)

AI/ML & GenAI	LLMs (Claude, Bedrock, OpenRouter), Embeddings, RAG, LangChain, Prompt Engineering, NLP, PyTorch, TensorFlow, MediaPipe, Computer Vision, Model Fine-tuning
Databases & Storage	PostgreSQL, MySQL, Redis, DynamoDB, FAISS (Vector DB), NoSQL, Graph DB Concepts, SQLAlchemy, database ORM
Cloud & DevOps	AWS (Lambda, S3, Bedrock, EC2, VPC, CloudFront), Docker, Jenkins, GitHub Actions, Terraform (basic), Linux, CI/CD, GCP Certified, Kubernetes (basic)
Big Data & Reliability	Kafka, Data Pipelines, Logging, Tracing, Observability, Scalability, Performance Tuning, A/B Testing
Software Practices	Agile (SCRUM), SDLC, Secure API Design, Responsible AI, Version Control (Git), Cross-functional Collaboration

PROJECTS

RAG-Powered PDF Chatbot | *Source Code* | *FastAPI, AWS Bedrock, Titan Embeddings, FAISS, Claude Sonnet, NLP* **Aug 2025**

- Built an end-to-end Retrieval-Augmented Generation (**RAG**) pipeline that ingests PDFs, splits text into semantic chunks, and embeds them using **AWS Titan** models; stored and queried efficiently through **FAISS** for high-recall context retrieval.
- Integrated Anthropic Claude Sonnet via **AWS Bedrock** with a **FastAPI** backend to deliver detailed, Markdown-formatted answers; implemented REST APIs, Swagger documentation, and persistent vector stores to showcase an enterprise-ready **GenAI** solution.

Github Pull Request Reviewer | *Article* | *React.js (Redux), Node.js (Express), REST, OAuth, HTML, TailwindCSS, NLP* **Feb 2025**

- Built reusable, backend-integrated AI-powered PR review system using Node.js, OpenRouter LLMs, and RESTful API design to assess code quality; enforced clean code practices, reduced manual review overhead by 40%, and improved engineering workflow.
- Designed secure, stateless OAuth-based authentication using React.js and Express; improved frontend performance, applied scalable CI/CD concepts with Docker, and debugged using profilers, optimizing review turnaround and UI responsiveness.

Enhanced Multimodal Hand Gesture Recognition | *Source Code* | *Pytorch, CNN, OpenCV, Streamlit, Matplotlib* **May 2024**

- Architected a hybrid fusion Convolutional Neural Network model using PyTorch, integrating multimodal inputs, data augmentation, and adaptive learning techniques, resulting in 94% accuracy for real-time gesture recognition.
- Deployed a Streamline-based application for real-time gesture prediction visualization, while managing version control, conducting code reviews, and driving SCRUM-based agile development for efficient project execution.

Thread Management & Scheduling in PintOS | *Source Code* | *C/C++, OS Scheduling, Synchronization* **Apr 2024**

- Designed and implemented priority-based scheduling in PintOS, optimizing thread wake-up and reducing CPU overhead through efficient sleep queue management, ensuring system stability in multi-threaded environments.
- Developed and tested priority donation mechanisms to prevent priority inversion, leveraging synchronization techniques and race condition mitigation for fault-tolerant OS scheduling.